3.46 (a) $z(t) = x(t)\,y(t) = \left(\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}\right)\left(\sum_{k=-\infty}^{+\infty} b_k e^{jk\omega_0 t}\right)$

$$= \sum_{k,l\in\mathbb{Z}} a_k b_l e^{j(k+l)\omega_0 t}$$

$$= \sum_{k=-\infty}^{+\infty}\sum_{n=-\infty}^{+\infty} a_n b_{k-n} e^{jk\omega_0 t}$$

$$\Rightarrow C_k = \sum_{n=-\infty}^{+\infty} a_n b_{k-n}$$

(b)

$$x_1(t) = \begin{cases} \cos 20\pi t, & -1\le t\le 1 \\ 0, & 1<t<2, \end{cases} \quad T_0 = 3$$

$$a_k = \frac{1}{T_0}\int_{-1}^{2} x_1(t)\, e^{-jk\omega_0 t}\,dt = \frac{1}{6}\int_{-1}^{1}\left(e^{j20\pi t} + e^{j(-20\pi t)}\right) e^{-jk\left(\frac{2\pi k}{3}\right)t}\,dt$$

$$= \frac{1}{6}\int_{-1}^{1}\left(e^{j\left(20\pi-\frac{2\pi k}{3}\right)t} + e^{j\left(-20\pi-\frac{2\pi k}{3}\right)t}\right)dt$$

$$= \frac{1}{6}\left(\frac{1}{j\left(20\pi-\frac{2\pi k}{3}\right)}\left(e^{j\left(20\pi-\frac{2\pi k}{3}\right)} - e^{j\left(-20\pi+\frac{2\pi k}{3}\right)}\right) + \frac{1}{j\left(20\pi-\frac{2\pi k}{3}\right)}\left(e^{j\left(-20\pi-\frac{2\pi k}{3}\right)} - e^{j\left(20\pi+\frac{2\pi k}{3}\right)}\right)\right)$$

$$= \frac{1}{3}\left(\frac{\sin\left(20\pi-\frac{2\pi k}{3}\right)}{20\pi-\frac{2\pi k}{3}} + \frac{\sin\left(20\pi+\frac{2\pi k}{3}\right)}{20\pi+\frac{2\pi k}{3}}\right)$$

$$= \frac{1}{3}\left(\mathrm{sinc}\left(20\pi-\frac{2\pi k}{3}\right) + \mathrm{sinc}\left(20\pi+\frac{2\pi k}{3}\right)\right)$$

$x_2(t) = z(t)\cos 20\pi t$.

$$\cancel{a_k = \frac{1}{3}\int_{-\infty}^{3} x_2(t)\, e^{-j\left(\frac{2\pi k}{3}\right)t}\,dt}$$

$$a_k = \frac{1}{3}\int_0^3 z(t)\, e^{-j\left(\frac{2\pi k}{3}\right)t}\,dt = \frac{1}{3}\left[\int_0^1 2e^{-j\left(\frac{2\pi k}{3}\right)t} + \int_1^2 e^{-j\left(\frac{2\pi k}{3}\right)t}\,dt\right]$$

$$= \frac{1}{3}\left(-\frac{1}{j\left(\frac{2\pi k}{3}\right)}\left(e^{-j\left(\frac{2\pi k}{3}\right)} + e^{-j\left(\frac{4\pi k}{3}\right)} - 2\right)\right)$$

$$= \frac{1}{3}\cdot\frac{1}{j\left(\frac{2\pi k}{3}\right)}\left(2 - 2\cos\left(\frac{2\pi k}{3}\right)\right)$$

$$C_k = \sum_{k=-\infty}^{+\infty}\sum_{n=-\infty}^{+\infty}\frac{1}{\pi n j}\left(1-\cos\left(\frac{2\pi n}{3}\right)\right)\cdot\frac{1}{3}\left[\mathrm{sinc}\left(20\pi-\frac{2\pi(k-n)}{3}\right)+\mathrm{sinc}\left(20\pi+\frac{2\pi(k-n)}{3}\right)\right]e^{-j\left(\frac{2\pi(k-n)}{3}\right)}$$

(e) $x_3(t) = \begin{cases} \frac{1}{2}\left(e^{-|t|+j20\pi t} + e^{-|t|-j20\pi t}\right), & |t| \le 1 \\ 0 & 1 \le |t| \le 2 \end{cases}, \quad T_0 = 4$

$$a_k = \frac{1}{4}\int_{-1}^{1} \frac{1}{2}\left(e^{-|t|+j\cdot 20\pi t} + e^{-|t|+j20\pi t}\right) dt$$

$$= \frac{1}{4}\int_0^1 \left(e^{(20\pi j-1)t} + e^{(-20\pi j-1)t}\right) dt$$

$$= \frac{1}{4}\left[\frac{e^{20\pi j-1}-1}{20\pi j-1} + \frac{e^{-20\pi j-1}-1}{-20\pi j-1}\right] = \frac{e^{-1}-1}{4}\left(\frac{1}{20\pi j-1} + \frac{1}{-20\pi j-1}\right) = \frac{1-e^{-1}}{4}\left(\frac{1}{1+20\pi j} - \frac{1}{1-20\pi j}\right)$$

$$= \frac{1-e^{-1}}{4} \cdot \frac{-40\pi j}{1+400\pi^2} =$$

$$a_k = \frac{1}{4}\int_{-1}^{1} \frac{1}{2}\left(e^{-|t|+j(20\pi t)} + e^{-|t|-j(20\pi t)}\right) e^{-j\left(\frac{2\pi k}{4}\right)t} dt =$$

$$= \frac{1}{4}\int_0^1 \left(e^{\left(\frac{39\pi}{2}j-1\right)t} + e^{\left(-\frac{41\pi}{2}j-1\right)t}\right) dt$$

$$= \frac{1}{4}\left(\frac{-e^{-1}j-1}{\frac{39}{2}\pi j-1} + \frac{e^{-1}j-1}{-\frac{41\pi}{2}j-1}\right)$$

$$= \frac{1}{8}\int_{-1}^{1}\left(e^{-|t|+j\left(20\pi-\frac{\pi k}{2}\right)t} + e^{-|t|-j\left(20\pi+\frac{\pi k}{2}\right)t}\right) dt$$

$$= \frac{1}{8}\int_{-1}^{0}\left[e^{\left[1+j\left(20\pi-\frac{\pi k}{2}\right)\right]t} + e^{\left[1-j\left(20\pi+\frac{\pi k}{2}\right)\right]t}\right] dt + \frac{1}{8}\int_0^1\left[e^{\left[-1+j\left(20\pi-\frac{\pi k}{2}\right)\right]t} + e^{\left[-1-j\left(20\pi+\frac{\pi k}{2}\right)\right]t}\right] dt$$

$$= \frac{1}{8}\left(\frac{1-e^{-1+j\frac{\pi k}{2}}}{1+j\left(20\pi-\frac{\pi k}{2}\right)} + \frac{1-e^{-1+j\frac{\pi k}{2}}}{1-j\left(20\pi+\frac{\pi k}{2}\right)}\right) + \frac{1}{8}\left(\frac{e^{-1+j\left(\frac{\pi k}{2}\right)}-1}{-1+j\left(20\pi-\frac{\pi k}{2}\right)} + \frac{e^{-1-j\left(\frac{\pi k}{2}\right)}-1}{-1-j\left(20\pi+\frac{\pi k}{2}\right)}\right).$$

(c) $y(t) = x^*(t), \quad b_k = a_k^* = a_{-k}$

$$|x(t)|^2 = x(t)\,y(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_0 t}$$

$$c_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\,y(t)\,dt = \frac{1}{T_0}\int_0^{T_0} |x(t)|^2 dt = \sum_{n=-\infty}^{+\infty} a_n b_{-n} = \sum_{n=-\infty}^{+\infty} a_n a_n^* = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

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$$b_k = \frac{1}{N}\sum_{n\in\langle N\rangle} y[n]\, e^{-j\left(\frac{2\pi k}{8}\right)n} = \frac{1}{N}\sum_{n\in\langle N\rangle}\left(\frac{1+(-1)^n}{2}\right) x[n-1]\, e^{-j\left(\frac{2\pi k}{8}\right)n}$$

$$a_k = \frac{1}{N}\sum_{n\in\langle N\rangle} x[n]\, e^{-j\left(\frac{2\pi k}{8}\right)n}$$

$$= \frac{1}{N}\sum_{n\in\langle N\rangle} x[n-1]\, e^{-j\left(\frac{2\pi k}{8}\right)n} \cdot e^{j\left(\frac{2\pi k}{8}\right)}$$

$$-a_{k-4} = -\frac{1}{N}\sum_{n\in\langle N\rangle} x[n]\, e^{-j\left(\frac{2\pi k}{8}\right)n} \cdot e^{j\pi n}$$

$$= \frac{1}{N}\sum_{n\in\langle N\rangle} x[n]\, e^{-j\left(\frac{2\pi k}{8}\right)n} (-1)^{n+1}$$

$$= \frac{1}{N}\sum_{n\in\langle N\rangle} x[n-1]\, e^{-j\left(\frac{2\pi k}{8}\right)n} \cdot e^{-j\left(\frac{2\pi k}{8}\right)} \cdot (-1)^n$$

$$b_k = \frac{a_k - a_{k-4}}{2} \cdot e^{\frac{j\pi k}{4}} = 2e^{\frac{j\pi k}{4}} a_k.$$

3.64 (a) $y(t) = \sum_{k=-\infty}^{+\infty} c_k \lambda_k \phi_k(t)$

(b) 系统是线性时变系统

(c) 令 $x(t) = \phi_k(t)$

$$y(t) = t^2 \cdot (k)(k-1)t^{k-2} + kt^{k-1} = k^2 t^k = k^2\phi_k(t) = k^2 x(t).$$

ϕ_k 是系统的 eigenfunction，$\lambda_k = k^2$

(d) $y(t) = 10\times(-10)^2 t^{-10} + 3\times 1^2 t + \frac{1}{2}\times 4^2 t^4 + 0^2\times\pi = 1000t^{-10} + 3t + 8t^4$